

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	26416	48.641084
1	[Git, Commit]	6744	12.418060
2	[Git, Change, Create]	5847	10.766370
3	[Git, Change, Delete]	4190	7.715254
4	[File, Git]	3576	6.584665
5	[Git, Change, Rename]	1699	3.128453
6	[Author, Git, Person]	1000	1.841349
7	[Git, Tag]	906	1.668263
8	[Json, Key]	668	1.230021
9	[Json, Value, Scalar]	603	1.110334
10	[Committer, Git, Person]	356	0.655520
11	[NPM, Dependency]	338	0.622376
12	[Type, TS, Primitive]	293	0.539515
13	[Type, TS, Declared]	277	0.510054
14	[TS, ExternalDeclaration]	215	0.395890
15	[Type, TS, Literal]	136	0.250424
16	[Json, Value, Object]	133	0.244899
17	[Type, TS, Union]	120	0.220962
18	[Type, TS, ObjectMember]	103	0.189659
19	[NPM, Script]	91	0.167563
20	[TS, Property]	65	0.119688
21	[TS, Function]	47	0.086543
22	[Type, TS, FunctionParameter]	40	0.073654
23	[Type, Object, TS]	39	0.071813
24	[File, Directory]	34	0.062606
25	[Type, TS, Function]	34	0.062606
26	[TS, Parameter]	33	0.060765
27	[Package, File, Json, NPM]	29	0.053399
28	[TS, Variable]	24	0.044192
29	[Value, TS, Literal]	20	0.036827

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

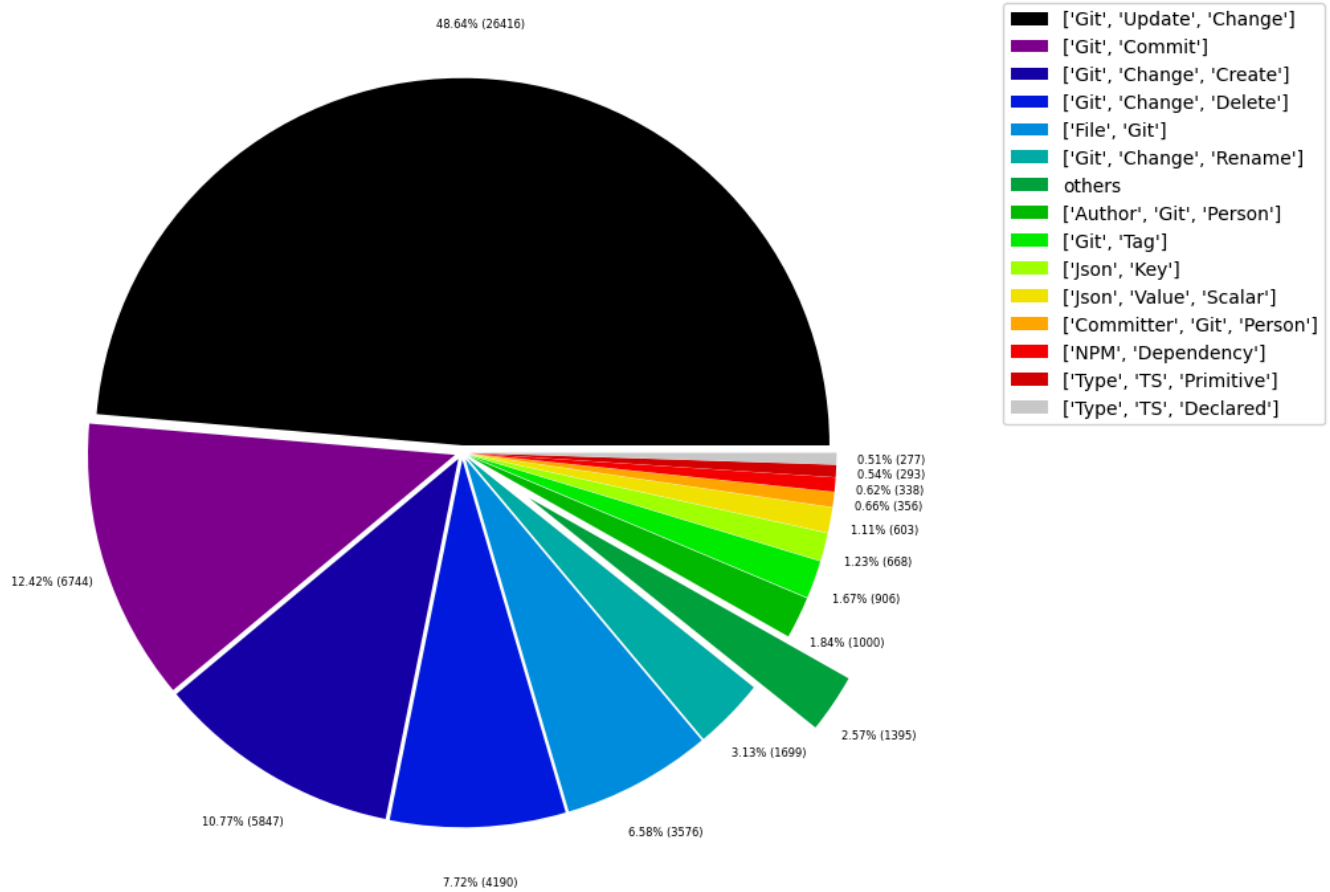


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.001841
1	[NPM, PublishConfig]	1	0.001841
2	[File, TS, Scan]	1	0.001841
3	[TS, Method]	1	0.001841
4	[Git, Branch]	1	0.001841
5	[TS, Constructor]	1	0.001841
6	[Value, TS, ObjectMember]	1	0.001841
7	[Repository, File, Git]	1	0.001841
8	[TS, Class]	1	0.001841
9	[TS, Enum]	2	0.003683
10	[Value, Object, TS]	3	0.005524
11	[Type, TS, Tuple]	3	0.005524
12	[Value, TS, Function]	4	0.007365
13	[TS, TypeParameter]	4	0.007365
14	[Value, TS, Complex]	5	0.009207
15	[Repository, NPM]	5	0.009207
16	[NPM, Engine]	6	0.011048
17	[Project, TS]	6	0.011048
18	[File, Local]	6	0.011048
19	[File, TS, Local, Module]	6	0.011048
20	[Value, TS, Call]	6	0.011048
21	[Type, TS, TypeParameterReference]	6	0.011048
22	[Value, TS, Member]	6	0.011048
23	[TS, EnumMember]	8	0.014731
24	[Type, TS, NotIdentified]	11	0.020255
25	[TS, ExternalModule]	11	0.020255
26	[Json, Value, Array]	12	0.022096
27	[Value, TS, Declared]	13	0.023938
28	[Git, Change, Copy]	15	0.027620
29	[TS, TypeAlias]	16	0.029462

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

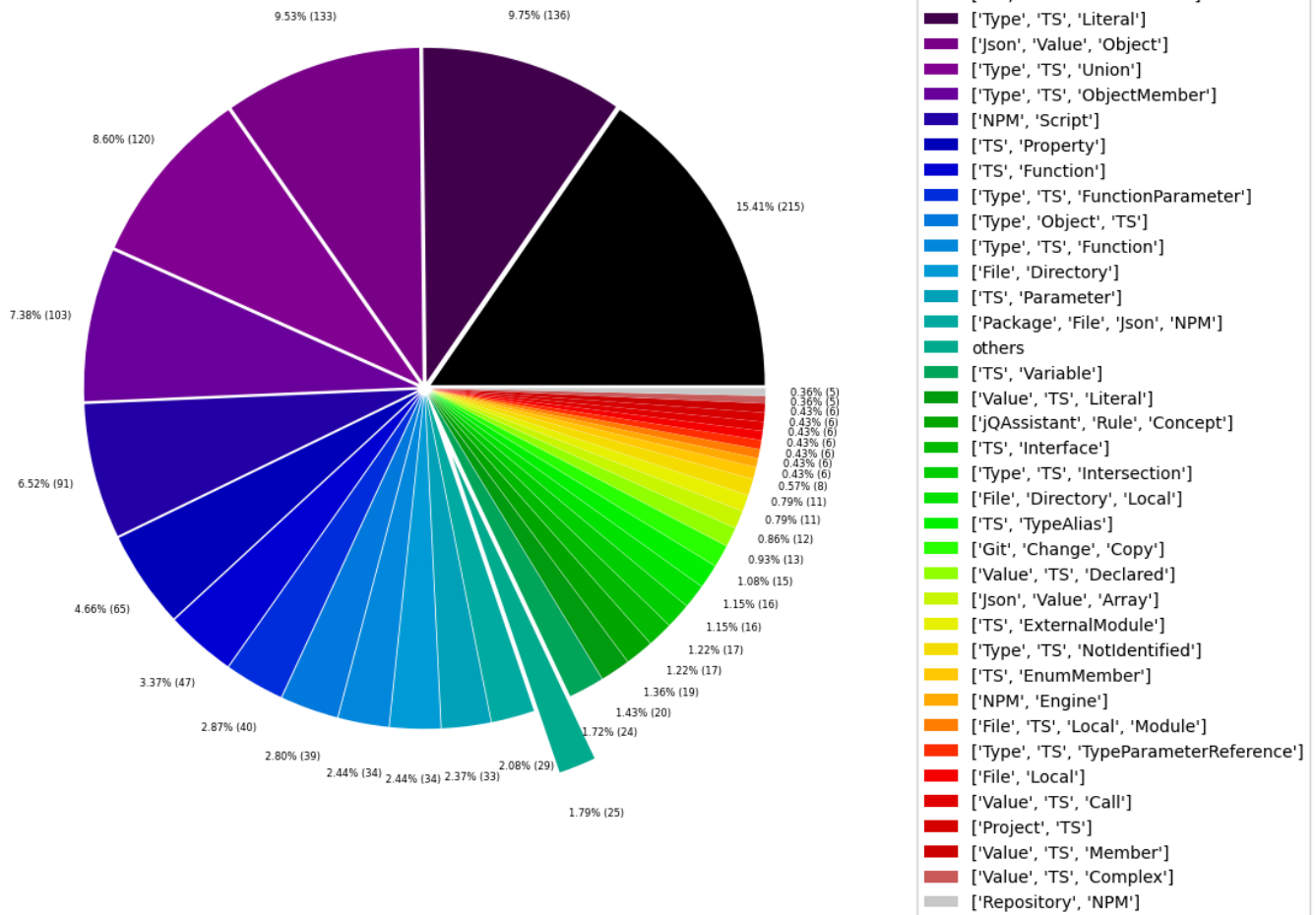


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

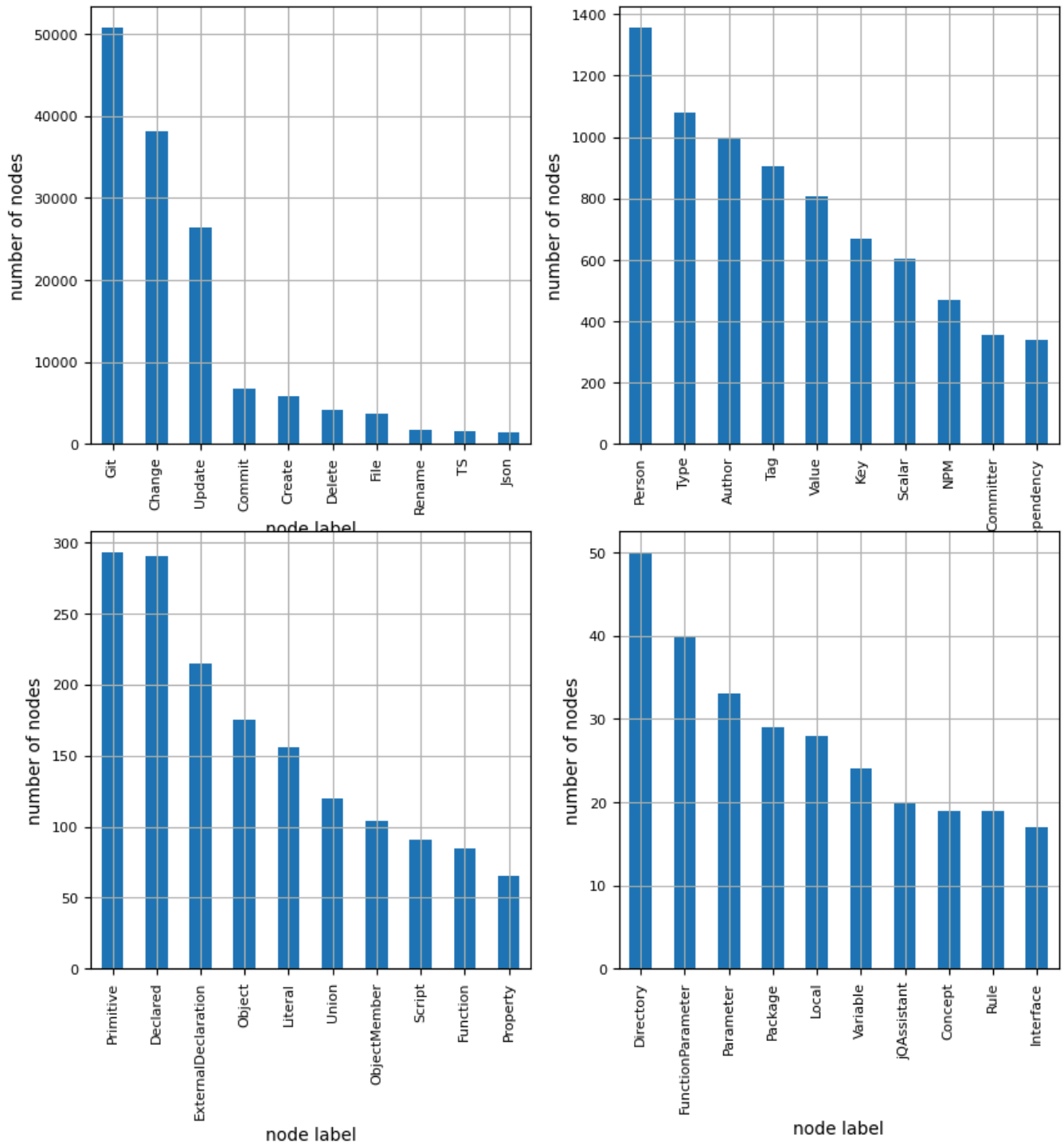
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	50751	93.450320
1	Change	38167	70.278780
2	Update	26416	48.641084
3	Commit	6744	12.418060
4	Create	5847	10.766370
5	Delete	4190	7.715254
6	File	3669	6.755911
7	Rename	1699	3.128453
8	TS	1595	2.936952
9	Json	1445	2.660750
10	Person	1356	2.496870
11	Type	1079	1.986816
12	Author	1000	1.841349
13	Tag	906	1.668263
14	Value	806	1.484128
15	Key	668	1.230021
16	Scalar	603	1.110334
17	NPM	470	0.865434
18	Committer	356	0.655520
19	Dependency	338	0.622376
20	Primitive	293	0.539515
21	Declared	290	0.533991
22	ExternalDeclaration	215	0.395890
23	Object	175	0.322236
24	Literal	156	0.287250
25	Union	120	0.220962
26	ObjectMember	104	0.191500
27	Script	91	0.167563
28	Function	85	0.156515
29	Property	65	0.119688
30	Directory	50	0.092067
31	FunctionParameter	40	0.073654
32	Parameter	33	0.060765
33	Package	29	0.053399
34	Local	28	0.051558
35	Variable	24	0.044192
36	jQAssistant	20	0.036827
37	Concept	19	0.034986
38	Rule	19	0.034986
39	Interface	17	0.031303

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 160177

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	38167	23.828015
1	MODIFIES	38167	23.828015
2	UPDATES	26416	16.491756
3	COMMITTED	13488	8.420685
4	CREATES	7561	4.720403
5	HAS_PARENT	7517	4.692933
6	HAS_COMMIT	6744	4.210342
7	DELETES	5889	3.676558
8	HAS_FILE	3576	2.232530
9	CHANGED_TOGETHER_WITH	1801	1.124381
10	RENAMES	1699	1.060702
11	HAS_NEW_NAME	1032	0.644287
12	HAS_AUTHOR	1000	0.624309
13	HAS_TAG	906	0.565624
14	ON_COMMIT	906	0.565624
15	DEPENDS_ON	887	0.553762
16	HAS_KEY	668	0.417039
17	HAS_VALUE	668	0.417039
18	CONTAINS	596	0.372088
19	HAS_COMMITTER	356	0.222254
20	OF_TYPE	339	0.211641
21	EXPORTS	309	0.192912
22	REFERENCES	197	0.122989
23	DECLARES	186	0.116122
24	DECLARES_DEV_DEPENDENCY	169	0.105508
25	DECLARES_DEPENDENCY	161	0.100514
26	HAS_MEMBER	104	0.064928
27	HAS_TYPE_ARGUMENT	94	0.058685
28	DECLARES_SCRIPT	91	0.056812
29	RETURNS	82	0.051193

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

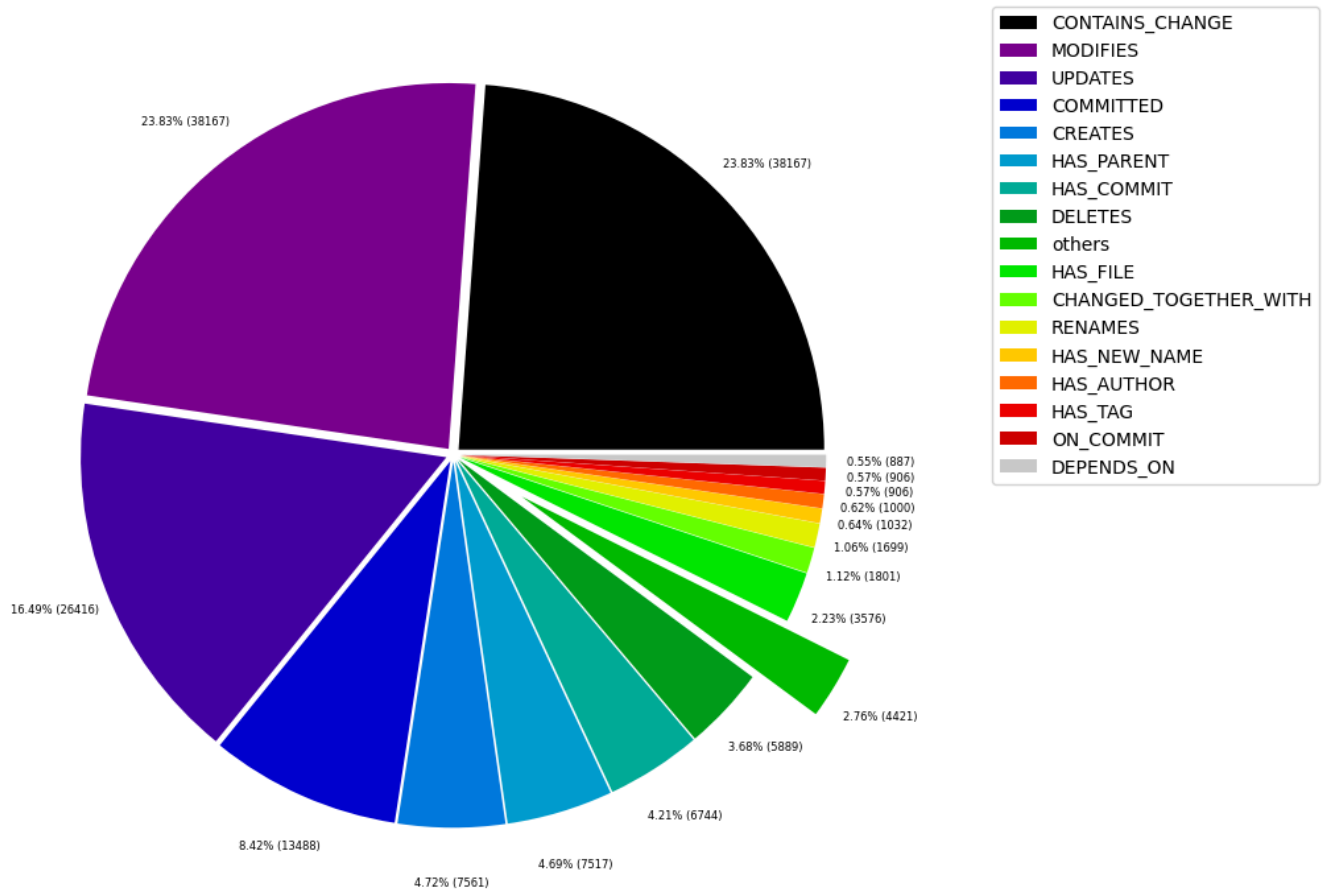


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

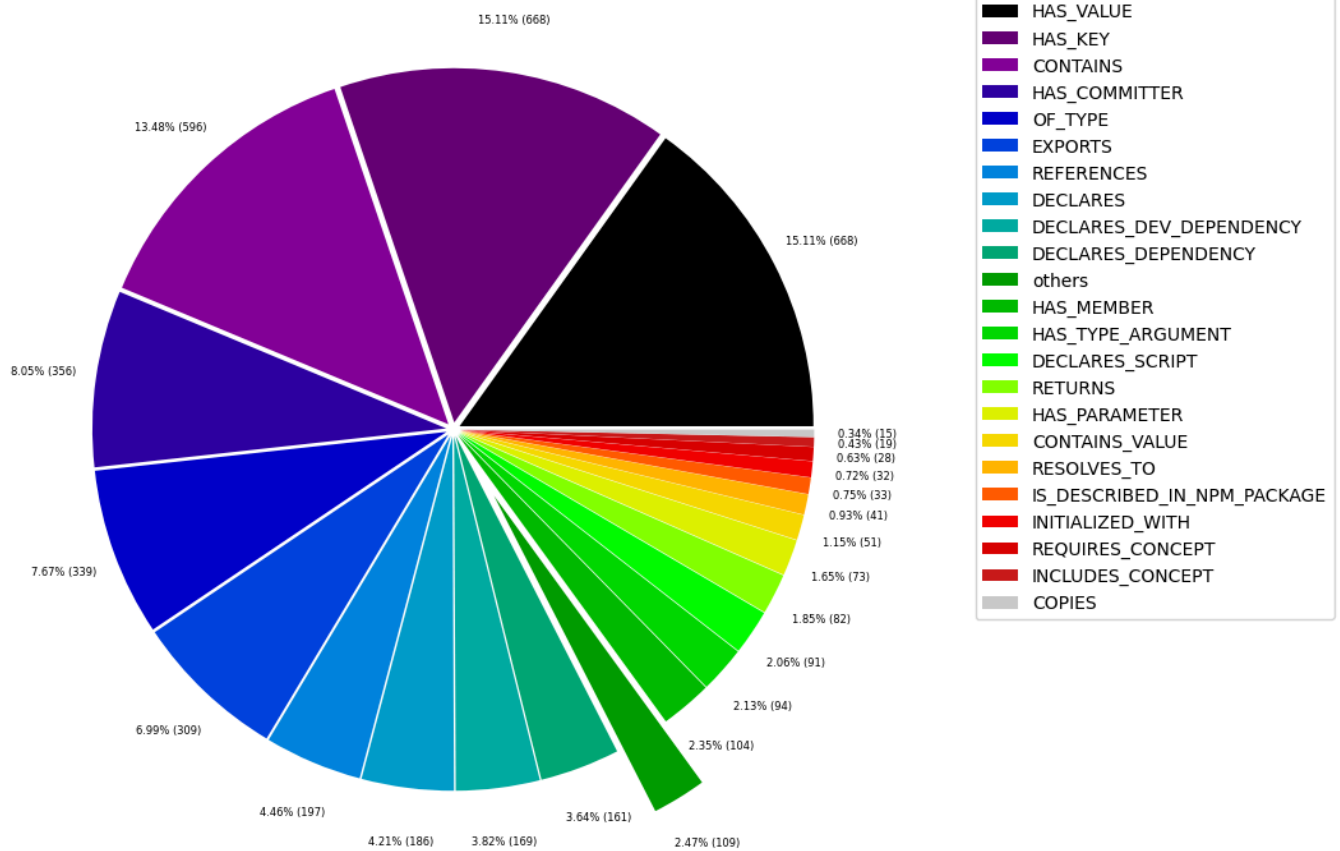
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_BRANCH	1	0.000624
1	DECLARES_PUBLISH_CONFIG	1	0.000624
2	HAS_HEAD	2	0.001249
3	CONSTRAINED_BY	4	0.002497
4	REFERENCED_PROJECTS	5	0.003122
5	IN_REPOSITORY	5	0.003122
6	DECLARES_ENGINE	6	0.003746
7	EXTENDS	6	0.003746
8	HAS_ARGUMENT	6	0.003746
9	CALLS	6	0.003746
10	HAS_NPM_PACKAGE	6	0.003746
11	HAS_CONFIG	6	0.003746
12	PARENT	6	0.003746
13	CONTAINS_PROJECT	6	0.003746
14	MEMBER	6	0.003746
15	HAS_ROOT	6	0.003746
16	DECLARES_PEER_DEPENDENCY	8	0.004994
17	USES	11	0.006867
18	COPY_OF	12	0.007492
19	COPIES	15	0.009365
20	INCLUDES_CONCEPT	19	0.011862
21	REQUIRES_CONCEPT	28	0.017481
22	INITIALIZED_WITH	32	0.019978
23	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.020602
24	RESOLVES_TO	41	0.025597
25	CONTAINS_VALUE	51	0.031840
26	HAS_PARAMETER	73	0.045575
27	RETURNS	82	0.051193
28	DECLARES_SCRIPT	91	0.056812
29	HAS_TYPE_ARGUMENT	94	0.058685

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOfNodesWithSameLabelsAsTarget
0	[Git, Update, Change]	MODIFIES	[File, Git]	26416	26416	
1	[Git, Update, Change]	UPDATES	[File, Git]	26416	26416	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	26416	6744	
3	[Git, Commit]	HAS_PARENT	[Git, Commit]	7517	6744	
4	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	6744	1	
5	[Author, Git, Person]	COMMITTED	[Git, Commit]	6744	1000	
6	[Committer, Git, Person]	COMMITTED	[Git, Commit]	6744	356	
7	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	5847	6744	
8	[Git, Change, Create]	MODIFIES	[File, Git]	5847	5847	
9	[Git, Change, Create]	CREATES	[File, Git]	5847	5847	
10	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	4190	6744	
11	[Git, Change, Delete]	DELETES	[File, Git]	4190	4190	
12	[Git, Change, Delete]	MODIFIES	[File, Git]	4190	4190	
13	[Repository, File, Git]	HAS_FILE	[File, Git]	3576	1	
14	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	1699	6744	
15	[Git, Change, Rename]	MODIFIES	[File, Git]	1699	1699	
16	[Git, Change, Rename]	RENAMES	[File, Git]	1699	1699	
17	[Git, Change, Rename]	CREATES	[File, Git]	1699	1699	
18	[Git, Change, Rename]	DELETES	[File, Git]	1699	1699	
19	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	1648	3576	
20	[File, Git]	HAS_NEW_NAME	[File, Git]	1032	3576	
21	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1000	1	
22	[Repository, File, Git]	HAS_TAG	[Git, Tag]	906	1	
23	[Git, Tag]	ON_COMMIT	[Git, Commit]	906	906	
24	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
25	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
26	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	356	1	
27	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
28	[File, TS, Local, Module]	DEPENDS_ON	[TS, ExternalDeclaration]	236	6	
29	[TS, ExternalModule]	EXPORTS	[TS, ExternalDeclaration]	215	11	

Graph Density

total_number_of_nodes (vertices): 54308

total_number_of_relationships (edges): 160177

-> total directed graph density: 5.431009139905184e-05

-> total directed graph density in percent: 0.005431009139905184